

Red flags in MSK conditions

Urgent and emergency musculoskeletal conditions requiring referral (NHSE 2020) and other references in text

Red Whale



GEMS

Guidelines & Evidence Made Simple

Condition	When to suspect and what to do!
Acute spinal cord compression	The spinal cord itself is compressed, usually by trauma/cancer/abscess or haematoma. Can occur up to any level, but remember that spinal cord ends at L1/2 (NEJM 2017; 376:1358). Consider if: - Weakness. - Lower motor neurone pattern AT THE LEVEL of the lesion = EVERYTHING goes down! (reduced tone, down-going plantars, reduced/absent reflexes) - Upper motor neurone signs BELOW the level of the lesion = EVERYTHING goes up! (increased tone, up-going plantars, brisk reflexes) - New-onset urinary incontinence or retention. - May have a sensory level BUT THIS IS A LATE SIGN – if you suspect and this is absent, still refer! Refer immediately for CT/MRI and neurosurgery assessment.
Cauda equina syndrome	Same process as acute spinal cord compression, but occurs in the cauda equina (usually below L1/L2). Central disc prolapse is most common cause, but can be trauma, cancer, abscess or haematoma. Look for: - Weakness. - Lower motor neurone pattern = EVERYTHING goes down (↓ tone, ↓ plantars, reduced/absent reflexes). - Saddle anaesthesia/reduced anal tone. CES symptoms <2 weeks: refer immediately for CT/MRI and neurosurgery assessment. CES symptoms ≥2 weeks OR bilateral sciatica without CES symptoms: urgent referral to MSK service.
Infection, e.g. discitis, osteomyelitis, septic arthritis or abscess	Relatively uncommon. Consider in IVDU or long-term steroids/immunosuppression, history of TB or children with an unexplained new limp. Note that osteomyelitis may be found in any bone and septic arthritis in any joint. Consider in adults or children if: - Fever/malaise/systemically unwell. - Increasing and unremitting pain; may be aggravated by straining. - May have tenderness over the vertebral body or affected bony area. Refer immediately to secondary care.
Trauma	If you are asked to assess a patient with a history of <u>traumatic</u> back pain, we can use the NICE 2016 NG41 guidance to identify who needs urgent imaging and immobilisation; most of us will need to call the paramedics to achieve this. NICE identifies high-risk features as: - Age 65y or older and reporting thoracic/lumbosacral pain. - High-risk injury, e.g. fall from height >3m, high-speed or roll-over motor collision, ejection from a motor vehicle, axial load injury (e.g. falling onto feet/buttocks), bicycle or horse-riding accidents. - Known or high risk of osteoporosis. - Abnormal examination findings, e.g. neurological signs, new deformity or midline tenderness, spinal pain on coughing. - Pain or abnormal neurology when mobilising (stop mobilising immediately if this occurs!). Refer immediately via ambulance (for imaging) if any of these features are present. If you need to acutely assess a cervical spine, NICE suggests using the Canadian C-Spine rules.
Acute limp in children	A new, unexplained acute limp in a child needs careful consideration. Possible explanations include serious causes such as septic arthritis, non-accidental injury, slipped upper femoral epiphysis (SUFE), Perthe's disease; more benign causes include transient synovitis (BMJ 2010;341:c4250). Refer for same-day assessment in secondary care children with an acute unexplained limp who are any of: - Aged <3y. - Unable to weight-bear. - Febrile or systemically unwell. - Aged >9y and reporting hip pain or restricted hip movement (to rule out SUFE). For a child aged 3y to 9y who is well, afebrile, mobile but limping for less than 48h, a short period of observation is reasonable because transient synovitis is most common in this age group. We should: - Reassess after 48h: if symptoms are resolving, a diagnosis of transient synovitis can be made without further investigation. - Safety-net carefully: if symptoms worsen or the child develops fever or systemic symptoms, urgent reassessment is required. If the symptoms worsen or fail to resolve after 7d, start investigations.

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Suspected inflammatory arthritis or acute rheumatological condition	Consider the diagnosis based on the history: - Joint pains and synovitis. - Early morning stiffness lasting >30 minutes and relieved by exercise. - Fatigue and systemic upset. In the presence of a good history, remember that up to 35% may have NORMAL CRP and ESR, and 30% will have normal anti-CCP/RhF – this should not put us off referring. Refer urgently to secondary care.		
Giant cell arteritis	Consider if age >50y and (BMJ 2019;365:l1964): - Age at onset >50y. - New-onset headache (usually temporal but can be parietal or occipital). - High ESR (>50mm/h). - Abnormal temporal artery on palpation. - Changes consistent with GCA on biopsy (not useful in primary care!). If 3 out of 5 are present, sensitivity is 93% and specificity is 91%. Other features may include visual disturbance/diplopia, scalp tenderness when brushing hair, pain in jaw on chewing or proximal muscle pain/stiffness (<i>may suggest co-existing polymyalgia rheumatica</i>). If typical history and visual disturbance present, needs <u>same-day referral</u> and ophthalmology assessment. If no visual disturbance, commence high-dose steroids in primary care and refer urgently to outpatients for confirmation of diagnosis (may want to discuss with secondary care at this time). (See <i>Giant cell arteritis</i> article.)		
Inflammatory back pain: axial spondyloarthritis	Suspect inflammatory back pain in people with low back pain lasting more than 3m, starting before the age of 45y AND with <u>4 or more</u> of these criteria (NICE 2017, NG65): - Onset before age 35y. - Waking in the second half of the night with symptoms. - Buttock pain. - Improvement with movement. - Improvement within 48h of taking NSAIDs. - First-degree relative with spondyloarthritis. - Current or past enthesitis. - Current or past psoriasis. If 4 or more criteria present, refer to rheumatology (no further testing required). If 3 criteria are present, do an HLA-B27 test and refer to rheumatology if positive. If fewer than 3 of these criteria are met, advise patients to re-present if they have new symptoms, particularly if they have a past or current history of inflammatory bowel disease, psoriasis or uveitis.		
Safeguarding	Any suspicion of non-accidental injury should be treated as at any other time, and be assessed and investigated urgently.		
Sarcoma (bone and soft tissue) NICE NG12, 2015	Clinical scenario	Adults	Children
	Unexplained bone swelling/pain	Urgent X-ray within 48h	Urgent X-ray within 48h
	Unexplained soft tissue lump that is enlarging	Urgent USS within 2 weeks	Urgent USS within 48h
	X-ray suggestive of bone sarcoma or USS suggestive of soft tissue sarcoma	Refer on suspected cancer pathway	Refer to be seen within 48h

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. September 2023. For full references see the relevant Red Whale articles.